MINING DEALER BEST PRACTICE

Performing Condition Monitoring Management Process to Save Component Life in Lift Cylinders

Maintenance and Repair	Application	**Component Life Management**	Component Renewal (CRC)	MARC Management

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1.0 Introduction

Manufacturers/dealers are constantly approached by customers on how to consistently achieve the best results in anticipating component failures using the condition monitoring process. In certain mining assets components (Ex: Diesel Engines and Transmission) used in mobile equipment models, this measurement can be done directly through “VIMs System” online from the online monitoring system data.

In other types of components like Cylinders, Pumps and small components it is not possible to detect failures using this path due to having few sensitive parameters to monitor.

Hence the CMMP (Condition Monitoring Management Process) can be improved by other practices to evaluate parameters behavior and anticipate failures. Hydro Paragominas, one of the most important customers in the Brazil North, developed and performed this process with good results.

Hydro Paragominas is proud of the work done in the development of this solution and wants to share the solution with the world. Hydro and Sotreq are willing to talk directly with other Caterpillar Dealers; the Sotreq contacts listed below will help facilitate the contact with Hydro.

2.0 Best Practice Description

This best practice is part of Hydro’s and Sotreq’s effort to improve the monitoring and troubleshooting process for cylinders based on the real case of the CMMP application.

The development of alternative monitoring/troubleshooting was based on the customer’s need to reduce the time to fix and improve fleet availability/productivity.

Thermography, and other operational tests, deliver the means to monitor and provide detection capabilities before catastrophic failure and helped define future intervention in the HEX, based on the HEX monitoring of components in the tool cylinder of a 390F.

Basic Operation:

There is literature that describes the operational verification (KENR9995-07), cylinder drop - verification - empty bucket (KENR9995-00) and cycle speed - verification (KENR9995-02) to evaluate the lifting cylinders of the boom.

Figure 1. Fluid Temperature

The equipment was positioned to perform the cylinder drop test (Figure 2), and the boom lift cylinder was measured from the rod fixing pin to the cylinder liner fixing pin, and after 5 minutes with the engine off, no displacement of the cylinder rod was identified through the measurement (the cylinders kept the same measurements).

Figure 2. Position of the drop test equipment

Before performing the cycle speed test, the pump relief pressures were checked for the boom lift function and found to be 5206 psi for the 01# pump and 5173 psi for the 02# pump (Figure 3).

Figure 3. Pump Pressure

Note: The pressures were a little above the specified value, but not enough to change the test result.

The machine implement was positioned as oriented by CAT literature to perform the cylinder cycle speed test and the cylinder cycle time from soil to fully extended rod position, and it was collected in 7.26 seconds.

The cylinder speed procedure specifies a time of 5.8 ± 0.5 for a new cylinder, a maximum of 7.5 seconds for reconditioning the cylinder and a maximum of 8.7 seconds for the service limit for the boom cylinder (Figure 4).

"Cylinder Speed - Check"					
Item	**Cylinder Position**	**Cylinder Operating Speed (seconds)**			
		New	**Rebuild(1)**	**Service Limit(1)**	**Actual**
Boom(2)	Extension	5.8 ± 0.5	7.5	8.7	
	Retraction	3.0 ± 0.3	3.9	4.5	
Boom(3)	Extension	6.0 ± 0.5	8.5	9.0	
	Retraction	3.1± 0.3	3.9	4.7	
Stick	Extension	4.7 ± 0.3	6.4	7.1	
	Retraction	3.7 ± 0.3	5.2	5.6	
Bucket(4)	Extension	4.2 ± 0.3	5.3	6.3	
	Retraction	2.5 ± 0.3	2.9	3.8	
Bucket(5)	Extension	4.9 ± 0.3	6.1	7.4	
	Retraction	3.1 ± 0.3	3.3	4.7	

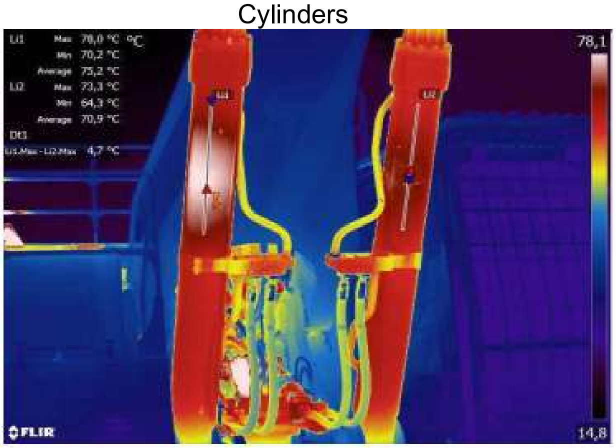
Figure 4. Pump Pressure Test

The procedure for setting the cycle time defined in Troubleshooting CAT was carried out, and the lifting time specified by the lower operation and maintenance manual was detected.

After finishing the cycle time and pressure tests, the thermography test was performed, with the aid of the thermographic camera, so that the differential temperature of the cylinders was recorded.

Using thermography on the EH3201, a thermal difference was identified in the boom cylinder on the right side, as can be seen in Figure 5a with a temperature differential of ~5°C in relation to the left side.

Figure 5a. Thermography on the EH3201 Lift Cylinders



Source: Field Technicians

Figure 5b. EH3201 on the operation front being inspected by Hydro's predictive team

Source: sis.cat.com

The hottest points of the cylinders were captured by the camera and it was observed that the lifting cylinder l/d of the boom has a temperature of 5°C above the lifting cylinder l/e (Figure 6).

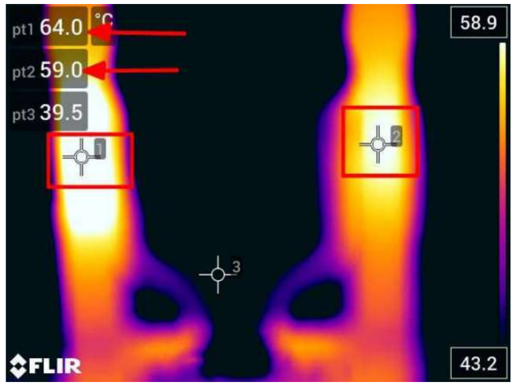
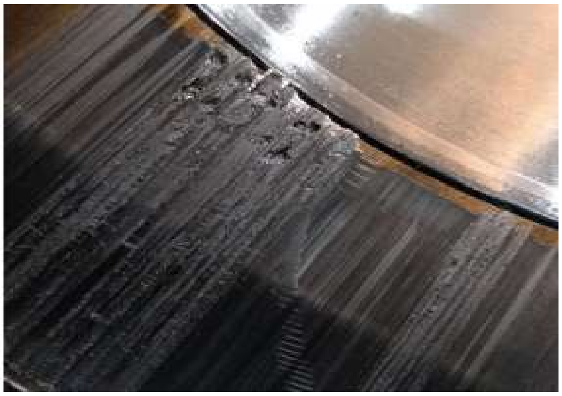


Figure 6. Thermographic image indicating ~5°C difference between the hottest points of the boom lift cylinders

The Remanufacturing Center (Cat-Dealer) evaluation of the cylinder allowed us to relate the field symptoms and conclude that the thermographic method can be used to define with more assurance the cylinder needing to be removed.

Figure 7. Cyclinder at the CRC

The next image shows some important details after disassembling the cylinder in the remanufacture center and evaluating internal evidence, confirming the relation between cylinder failure mode presented in the field and causes. See below that the failure mode presented in Figure 8 represents the cause of temperature increase in the field thermographic process.



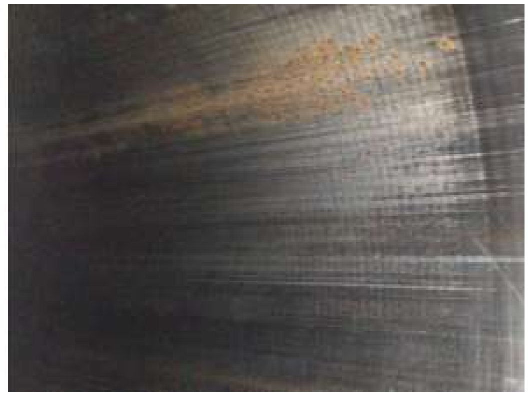


Figure 8. Internal failure at cylinder

Based on this study, it was possible to see the effectiveness of using this process to indicate possible internal oil leakage by the cylinder seal assembly. The Remanufacturing Center confirmed that removing the cylinder based on the thermography along with the other tests was shown to be the proper decision to save money.

3.0 Implementation Steps

1. Define and establish the right process to measurement
   - Operational Verification (KENR9995-07) – standard practice
   - Cylinder Drop Verification (KENR9995-00) – standard practice
   - Cycle Speed Verification (KENR9995-02) – standard practice
   - Thermography – added practice (key to this best practice)

   <u>Note: Please see the basic description in the above section for the detail.</u>

2. Apply the process in one real case in the field to define and remove components before failure

3. Send component to the CRC to evaluate internal failure stage and relation of failure to field failure mode (temperature and time to cycle)

4. Standardize the process to additional machines and cylinders

5. Continuously improve the process and measure the cost-saving of the CMMP procedure

4.0 Benefits

- Improved machine availability and productivity;
  - Components like cylinders that have a few typical condition monitoring parameters can be better monitored with additional methods like thermography
  - Replacement plans (parts, scheduled downtime, etc.) can be put in place before failure of the component
- Allows greater agility in temperature assessment (lower MTTR);

- Reduces risks through less time spent in the asset's operating zone (Elimination of live-work);
- Provides greater sensitivity in decision making for asset shutdown and definition of the next cycle time measurement step;
- Provides greater predictability in the cylinder replacement planning and scheduling process.
  - Wear evaluation increase the accuracy of the replacement forecast
  - Improves material stock planning by using the hours actually operated as a parameter

5.0 Resources Required

Parts: Genuine CAT parts were used to facilitate the purchase by the customer; this will support future repair needs/replacement.

6.0 Supporting Attachments / References

Thermal camera

7.0 Related Best Practices

N/A

8.0 Acknowledgements

SOTREQ would like to thank the immense commitment of its Mining Engineering team to deliver the best fleet performance.

All best practices are developed prioritizing safety; it is the way our team works for our customers, partners and ourselves.

We would like to mention and thank Doug Friedrich for the close collaboration with our company. Doug has provided necessary insights and very important technical input for project development.

Thanks to The Customer for challenging the dealer to figure out this solution; we are glad that you are happy and satisfied with the final result.

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